

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCEC602 - CONCRETE TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which type of cement is best for marine structures?
(a) Portland pozzolana cement (b) Rapid hardening cement
(c) Sulphate resisting cement (d) Low heat cement
 - 2 The bulking of sand occurs due to:
(a) Moisture content (b) High temperature
(c) Excessive compaction (d) Chemical reaction
 - 3 The chemical compound responsible for early strength in cement is:
(a) Tricalcium silicate (b) Dicalcium silicate
(c) Tricalcium aluminate (d) Tetracalcium aluminoferrite
 - 4 Abrasion resistance of aggregates is tested using:
(a) Los Angeles machine (b) Slump test
(c) Compressive test (d) Impact test
 - 5 Coarse aggregates typically have a size greater than:
(a) 4.75 mm (b) 2.36 mm (c) 1.18 mm (d) 600 microns
 - 6 The specific gravity of aggregates generally ranges between:
(a) 2.5 to 3.0 (b) 1.0 to 2.0 (c) 3.5 to 4.0 (d) 4.5 to 5.0
 - 7 Excess water in concrete mix leads to:
(a) Lower strength (b) Higher durability
(c) Reduced setting time (d) Improved bond
 - 8 Which batching method provides higher accuracy?
(a) Weigh batching (b) Volume batching (c) Hand batching (d) Visual batching
 - 9 The process of mixing concrete ensures:
(a) Uniform distribution of materials (b) Faster setting time
(c) Lower workability (d) Higher curing time
 - 10 Workability of concrete is the measure of its:
(a) Ease of mixing, placing, and compaction (b) Strength
(c) Durability (d) Curing time

- 11 Non-destructive testing of concrete is used to:
- (a) Assess properties without damaging the structure
 - (b) Increase strength
 - (c) Enhance appearance
 - (d) Change mix proportions
- 12 Acceptance criteria for concrete are based on:
- (a) Compressive strength tests (b) Flexural strength tests
 - (c) Workability tests (d) All of the above

Q2. Solve the following.

- A) Describe the hydration of cement. Explain the hydration products formed during the setting and hardening process. 6
- B) Explain the different types of cement used in concrete construction. 6

Q3. Solve the following.

- A) Explain the phenomena of segregation and bleeding in fresh concrete. Discuss their causes and effects on concrete quality. 6
- B) Define workability of fresh concrete. Explain the factors affecting workability 6

Q4. Solve Any Two of the following.

- A) Classify admixtures used in concrete and explain their general functions and importance in modern concrete construction. 6
- B) Describe plasticizers and super-plasticizers and their mechanism of action 6
- C) explain role of pozzolanic admixtures in contributing to sustainability and improvement of the properties of concrete. 6

Q5. Solve Any Two of the following.

- A) Describe the effect of water–cement (W/C) ratio on the strength of concrete. 6
- B) Explain the durability of concrete with reference to minimum and maximum cement content. <https://www.batuonline.com> 6
- C) Explain the causes of deterioration and cracking in concrete in relation to sulphate attack, and alkali–aggregate reaction (AAR). 6

Q6. Solve Any Two of the following.

- A) Explain the concept of concrete mix design and the factors governing the mix design of concrete. 6
- B) Describe the factors causing variations in concrete quality. Explain the importance of field control in minimizing these variations. 6
- C) Describe non-destructive testing (NDT) of concrete. Explain commonly used NDT methods. 6

*** End ***